Snake Game Development Report

1. Project Overview

The objective of this project was to develop a modern Snake Game using HTML, CSS, and JavaScript with the assistance of AI-generated prompts. The game was designed to be responsive and playable on both desktop and mobile devices. Features include multiple difficulty levels, score tracking, high score storage, pause functionality, and touch controls.

Deployed Application Link

https://meek-crostata-1ef696.netlify.app/

2. Prompt Iterations Used During Development

Initial Prompt

"Create a simple Snake Game in a single HTML file."

Result

A basic snake game was generated with simple movement, food collection, and score tracking.

Second Prompt

"Improve the design and add three difficulty levels."

Result

Level selection (Easy, Medium, Hard) was added with different snake speeds.

Third Prompt

"Make the game mobile-friendly and add touch controls."

Result

Mobile buttons and responsive layout were implemented.

Fourth Prompt

"Improve the snake appearance and game interface."

Result

The UI was enhanced with neon colors, better styling, glowing effects, and improved score panels.

Fifth Prompt

"After game over, return to level selection screen instead of restarting automatically."

Result

The game flow was improved so users select a difficulty level each time they start a new game.

3. What Worked Well

- AI generated a functional Snake Game quickly.
- Difficulty levels were implemented successfully.
- Mobile controls worked effectively.
- High score storage using Local Storage functioned correctly.
- The game remained within a single HTML file as required.
- Responsive design allowed gameplay on both desktop and mobile devices.

4. What Did Not Work Well

- Initial snake appearance was too basic and looked like simple square blocks.
- Early versions lacked proper game flow after game over.
- Some UI elements required multiple iterations before achieving an attractive design.
- Touch controls needed adjustment to improve responsiveness on mobile devices.

5. How Prompts Were Improved Over Time

The prompts became progressively more specific.

Example 1

Initial Prompt:
"Create a snake game."

Improved Prompt:
"Create a responsive Snake Game in a single HTML file with three difficulty levels, mobile controls, and high-score tracking."

Example 2

Initial Prompt:
"Improve the design."

Improved Prompt:
"Use a modern neon theme, glowing effects, rounded snake segments, and responsive layouts."

Example 3

Initial Prompt:
"Add levels."

Improved Prompt:
"Implement Easy, Medium, and Hard modes with different snake speeds and allow users to select a level before each game."

By providing more detailed requirements, the generated code became more accurate and closer to the desired outcome.

6. Challenges Faced During Development

Challenge 1: Game Design

The initial game design appeared too simple and required several improvements to achieve a modern look.

Challenge 2: Mobile Compatibility

Ensuring the game worked correctly on mobile devices required adding touch controls and responsive layouts.

Challenge 3: Game Flow

Managing transitions between the menu, gameplay screen, and game-over screen required additional coding and testing.

Challenge 4: Snake Appearance

Creating a snake that visually resembled a traditional snake rather than simple blocks required design modifications.

Challenge 5: Single File Requirement

All HTML, CSS, and JavaScript needed to remain inside one file, making organization and maintenance more challenging.